

Industrial Internship Report on "CS Nexus-A Computer Science Educational Website Using HTML& CSS"

Prepared by
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Executive Summary

This report provides details of the Industrial Internship conducted by **upskill Campus** and **The IoT Academy** in collaboration with the industrial partner **UniConverge Technologies Pvt. Ltd. (UCT)**.

This internship was focused on a project/problem statement provided by UCT. We had to complete the project along with the report within a period of six weeks.

My project was **CS Nexus – A Computer Science Educational Website Using HTML & CSS**. CS Nexus is a multi-page educational website designed to provide structured learning resources for Computer Science students. It includes separate pages for important subjects such as Data Structures, Algorithms, Database Management Systems, Operating Systems, Computer Networks, Cloud Computing, Artificial Intelligence, Machine Learning, and Software Engineering. It also contains programming language pages for Java, Python, C++, HTML, CSS, and JavaScript.

This project helped me improve my knowledge of HTML, CSS, website layout design, page navigation, and content organization. It also gave me practical experience in building a complete multi-page website and managing the project using GitHub.

This internship gave me a very good opportunity to gain exposure to industrial style project development and to design and implement a real-world educational solution. It was an overall great experience to have this internship.

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1 Preface

The six-week Industrial Internship organized by **upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)** provided me with an excellent opportunity to gain practical exposure to the field of **Data Science and Machine Learning**. The internship was designed to enhance technical knowledge, problem-solving abilities, and practical skills through the development of a real-world project.

As part of this internship, I developed **CS Nexus – A Computer Science Educational Website Using HTML & CSS**. The objective of this project was to create a simple, informative, and user-friendly educational platform that provides learning resources on important Computer Science subjects and programming languages. The website includes dedicated pages for Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Cloud Computing, Artificial Intelligence, Machine Learning, Software Engineering, and programming languages such as Java, Python, C++, HTML, CSS, and JavaScript.

The internship was planned over six weeks with well-defined stages. Initially, I analyzed the project requirements and planned the overall structure of the website. In the following weeks, I designed the user interface, developed multiple interconnected webpages using HTML and CSS, tested all navigation links and layouts, organized the educational content, and finally documented the project and uploaded it to GitHub.

Although the internship domain was **Data Science and Machine Learning**, I selected a web development project that allowed me to strengthen my front-end development skills while also presenting educational content related to Data Science, Machine Learning, and other Computer Science subjects. This project helped me improve my knowledge of HTML5, CSS3, website design, content organization, GitHub, debugging, documentation, and project management.

I sincerely thank **upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)** for providing me with this valuable internship opportunity. I also express my gratitude to the mentors, trainers, and faculty members for their continuous guidance, encouragement, and support throughout the internship.

Finally, I encourage my juniors and fellow students to actively participate in internship programs like this. Such opportunities provide practical industry exposure, enhance technical knowledge, improve confidence, and prepare students for successful careers in technology.

Thank to all my friends, who have helped you directly or indirectly.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



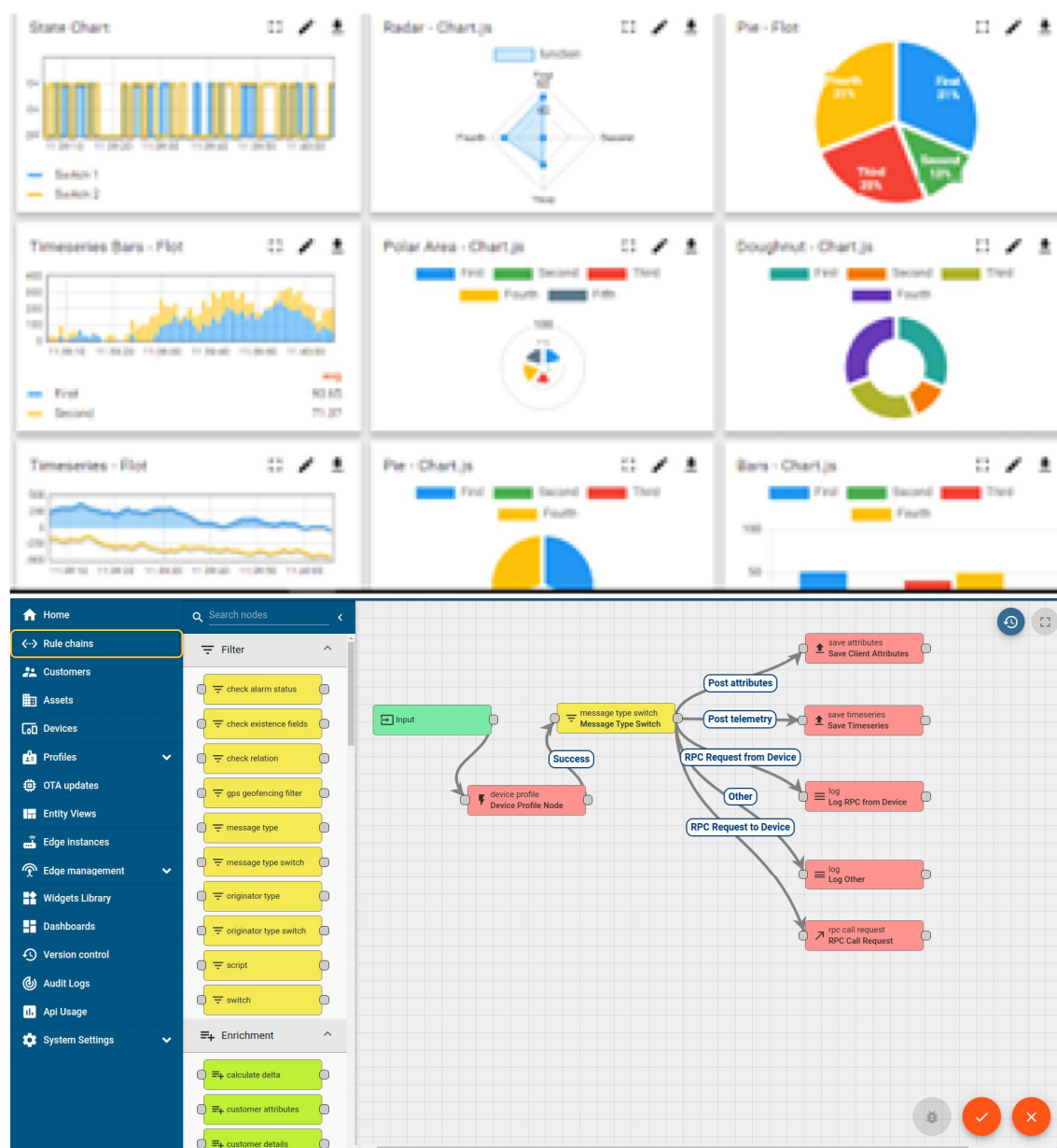
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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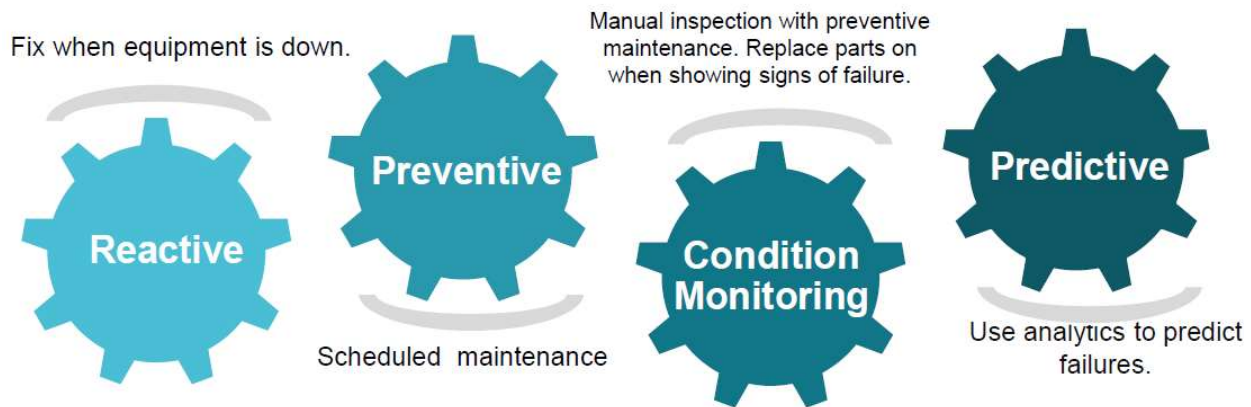


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

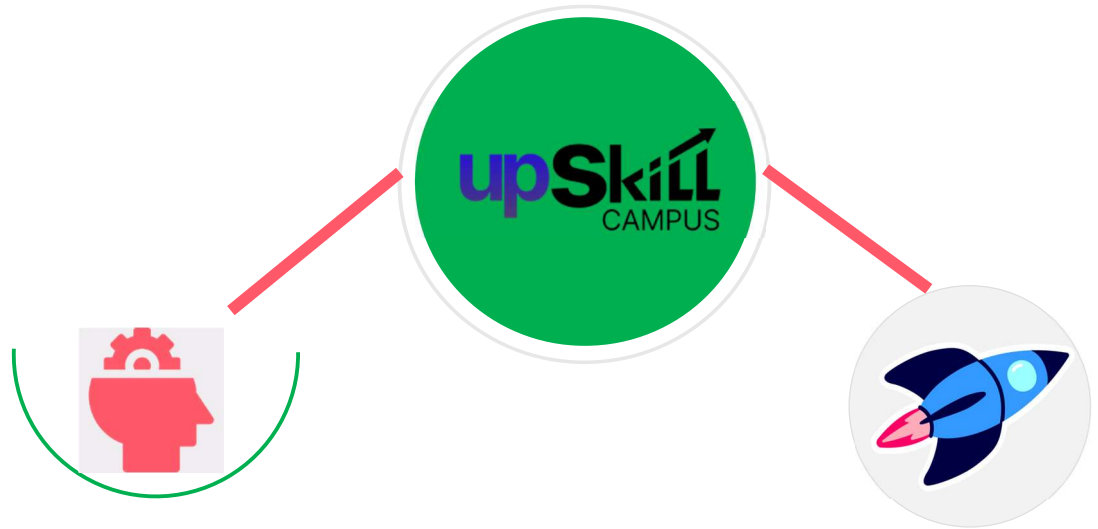
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

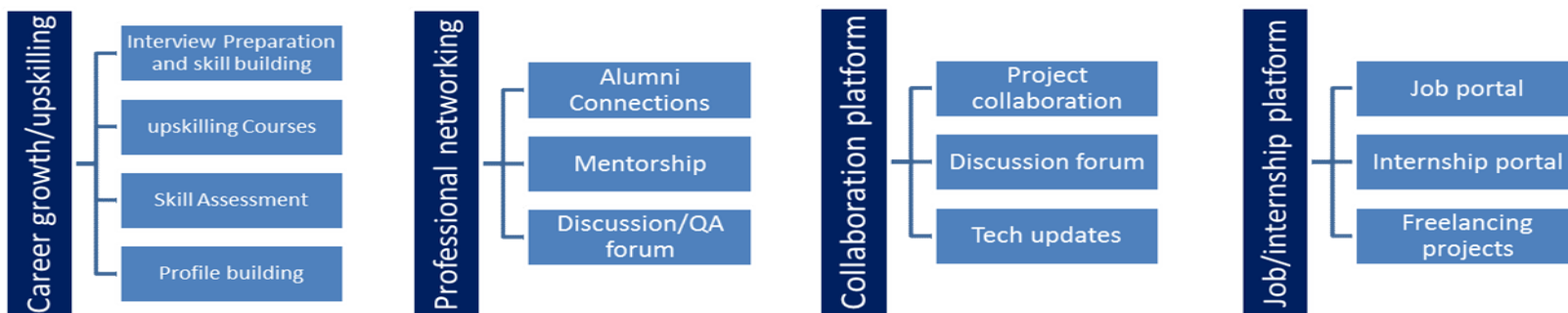
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] [2] W3Schools. *HTML and CSS Tutorials*. Available at: <https://www.w3schools.com/>
- [2] GitHub. *GitHub Documentation and Repository Hosting*. Available at: <https://github.com/>
- [3] Visual Studio Code. *Official Documentation*. Available at: <https://code.visualstudio.com/docs>

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Machine Learning	ML
Operating System	OS
Hyper Text Markup Language	HTML
Cascading Style Sheets	CSS

Problem Statement

3 3. Problem Statement

In today's digital era, students have access to a vast amount of educational content available across different websites and online platforms. However, this information is often scattered, making it difficult for beginners to find organized and reliable learning resources in one place. Students frequently switch between multiple websites to study Computer Science subjects and programming languages, which can lead to confusion and consume valuable time.

Many existing educational websites provide comprehensive information, but they may contain excessive advertisements, complex navigation, or advanced content that is not suitable for beginners. As a result, students often struggle to identify the most important concepts and learning paths.

To address this problem, the project **CS Nexus – A Computer Science Educational Website Using HTML & CSS** was developed. The objective of the project is to provide a simple, organized, and user-friendly platform where students can access essential information about various Computer Science subjects and programming languages in one place.

The website includes dedicated pages for subjects such as **Data Structures and Algorithms (DSA)**, **Database Management Systems (DBMS)**, **Operating Systems (OS)**, **Computer Networks (CN)**, **Cloud Computing**, **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Software Engineering**, along with programming languages including **Java**, **Python**, **C++**, **HTML**, **CSS**, and **JavaScript**. The website is designed with a clean interface, easy navigation, and responsive layout to enhance the learning experience.

By providing structured educational content in a single platform, CS Nexus aims to simplify the learning process, improve accessibility to fundamental Computer Science concepts, and help beginners build a strong foundation in technical subjects.

4 Existing and Proposed solution

• 4.1 Existing Solution

Today, numerous online platforms provide educational content related to Computer Science and programming. Popular websites such as **W3Schools**, **GeeksforGeeks**, **TutorialsPoint**, **MDN Web Docs**, and **freeCodeCamp** offer tutorials, examples, and technical documentation for learners. These platforms are widely used because they cover a broad range of topics and provide detailed explanations.

However, despite their advantages, these platforms have certain limitations for beginners:

- Learning resources are spread across multiple websites, making it difficult to find all topics in one place.
- Some websites contain excessive advertisements and distractions that affect the learning experience.
- Beginners may find the large amount of content overwhelming and may struggle to identify where to start.
- Navigation between different subjects is not always simple for new learners.
- Some advanced topics are explained without sufficient beginner-level guidance.

Because of these limitations, students often spend more time searching for information than actually learning the concepts.

• 4.2 Proposed Solution

To overcome these challenges, **CS Nexus – A Computer Science Educational Website Using HTML & CSS** was developed. The project provides a clean, organized, and beginner-friendly platform where students can access essential Computer Science learning resources from a single website.

The website contains dedicated pages for major Computer Science subjects, including **Data Structures and Algorithms**, **Database Management Systems**, **Operating Systems**, **Computer Networks**, **Cloud Computing**, **Artificial Intelligence**, **Machine Learning**, and **Software Engineering**. It also provides individual pages for programming languages such as **Java**, **Python**, **C++**, **HTML**, **CSS**, and **JavaScript**.

The website has been designed with a simple navigation system and an attractive user interface to make learning more accessible. Users can easily switch between different subjects and programming languages without navigating through multiple websites.

• Key Features of the Proposed Solution

- Simple and attractive user interface.
- Easy navigation between all webpages.
- Dedicated pages for individual subjects and programming languages.
- Beginner-friendly educational content.
- Responsive website layout using HTML and CSS.
- Organized learning resources in a single platform.
- Fast loading and lightweight webpages.
- Easy to maintain and extend with additional content in the future.
- **Value Addition**

The proposed solution adds value by providing students with a centralized educational platform where they can quickly access fundamental Computer Science topics without unnecessary distractions. It improves the learning experience through a structured website layout, consistent navigation, and easy-to-understand content. The project also serves as a foundation that can be expanded with interactive features, quizzes, video tutorials, and backend support in the future.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

The proposed system, **CS Nexus – A Computer Science Educational Website**, is designed as a static multi-page website developed using **HTML5** and **CSS3**. The website provides an organized platform where users can access information about Computer Science subjects and programming languages through a simple and intuitive navigation system.

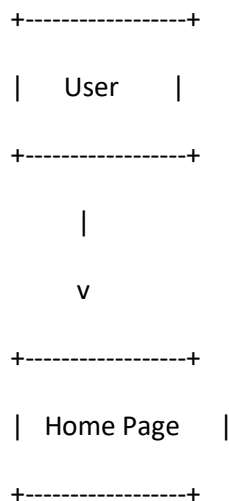
The design emphasizes ease of use, readability, and structured content presentation. Every webpage follows a consistent layout consisting of a navigation bar, content section, and footer, ensuring a uniform user experience throughout the website.

The project follows a modular design approach, where each subject and programming language has its own dedicated webpage. This makes the website easier to maintain, update, and expand in the future.

The overall workflow begins from the Home page, where users can navigate to different sections such as Subjects, Programming Languages, Roadmap, About, and Contact. Each subject or programming language page provides detailed information, allowing users to easily explore educational content without leaving the website.

The website was designed to be lightweight, responsive, and compatible with modern web browsers. The navigation links were carefully tested to ensure smooth transitions between all webpages.

5.1 High Level Diagram (if applicable)



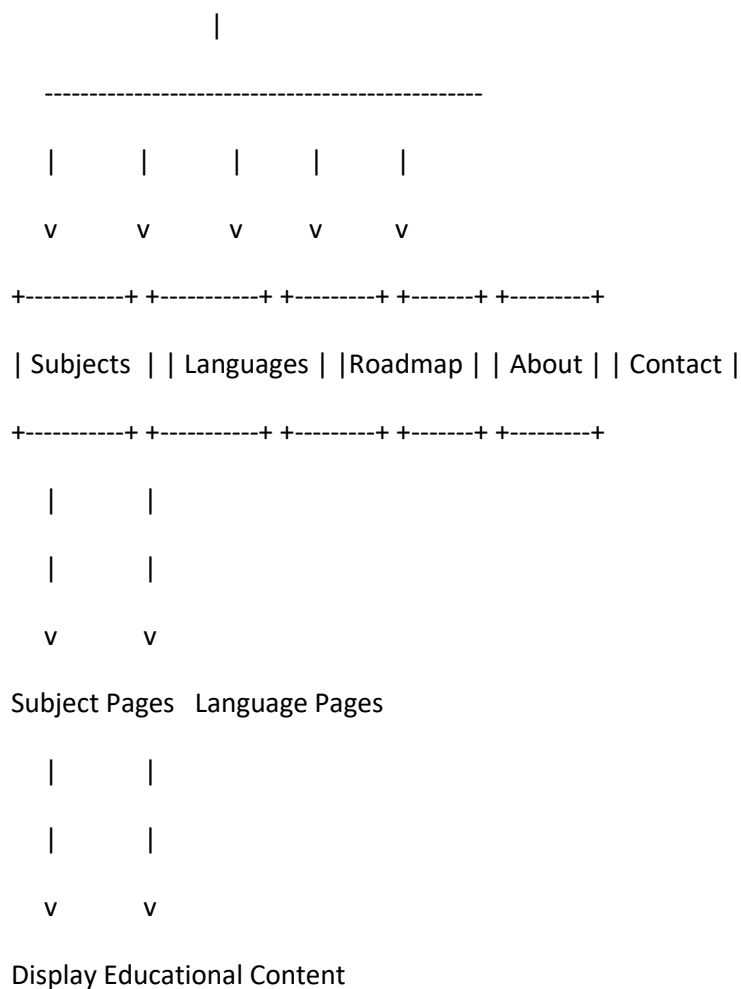


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

User Opens Website



Loads Home Page

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Navigation Menu

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Subjects Page

Languages Page

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Select Subject

Select Language

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Display Information

Display Information

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Back to Navigation

5.3 Interfaces (if applicable)

The user interface of **CS Nexus** is designed to be clean, responsive, and beginner-friendly. Every webpage follows a consistent design theme to provide a smooth browsing experience.

The website consists of the following interfaces:

- **Home Page**
 - Displays the project title and introduction.
 - Provides navigation to all major sections.
- **Subjects Page**
 - Lists all Computer Science subjects.
 - Each subject contains a "Learn More" button leading to a dedicated webpage.
- **Programming Languages Page**
 - Displays popular programming languages.
 - Each language includes an "Explore" button for detailed information.
- **Roadmap Page**
 - Provides a recommended learning path for beginners.
- **About Page**
 - Gives information about the project and its objectives.
- **Contact Page**
 - Displays contact details and feedback information.
- **Individual Subject Pages**
 - DSA
 - Algorithms

- DBMS
- Operating Systems
- Computer Networks
- Cloud Computing
- Artificial Intelligence
- Machine Learning
- Software Engineering
- **Individual Programming Language Pages**
- Java
- Python
- C++
- HTML
- CSS
- JavaScript

Insert screenshots here:

- Figure 3: Home Page
- Figure 4: Subjects Page
- Figure 5: Languages Page
- Figure 6: Roadmap Page
- Figure 7: About Page
- Figure 8: Contact Page

Performance Test

1. 6. Performance Test

Performance testing is an important stage in software development because it verifies that the developed system functions correctly under different conditions. Although **CS Nexus** is a static educational website developed using HTML and CSS, several tests were performed to ensure that the website provides a smooth and reliable user experience.

The website was tested for navigation, page loading, browser compatibility, responsive layout, and link functionality. Every webpage was checked to ensure that users could easily navigate between sections without encountering errors or broken links.

The following constraints were considered during the development of the website:

- Fast loading speed.
- Proper navigation between webpages.
- Responsive layout for different screen sizes.
- Consistent user interface across all pages.
- Lightweight website without unnecessary resources.
- Easy accessibility and readability of content.

Appropriate HTML structure and optimized CSS styling were used to minimize loading time and provide a consistent browsing experience. All hyperlinks and navigation buttons were tested to ensure correct redirection to their respective webpages.

2. 6.1 Test Plan / Test Cases

The following test cases were executed during project testing.

Test Case ID	Test Description	Expected Result	Status
TC-01	Open Home Page	Home page loads successfully	Passed

Test Case ID	Test Description	Expected Result	Status
TC-02	Open Subjects Page	Subjects page displays correctly	Passed
TC-03	Open Languages Page	Languages page displays correctly	Passed
TC-04	Click "Learn More" buttons	Opens corresponding subject page	Passed
TC-05	Click "Explore" buttons	Opens corresponding language page	Passed
TC-06	Navigation Menu	All navigation links work correctly	Passed
TC-07	About Page	Displays project information	Passed
TC-08	Contact Page	Displays contact information	Passed
TC-09	Browser Compatibility	Website works properly in Google Chrome and Microsoft Edge	Passed
TC-10	Responsive Layout	Website adjusts properly on different screen sizes	Passed

3. 6.2 Test Procedure

The website was tested manually after completing the development process.

The following testing procedure was followed:

4. Opened the website using **Google Chrome**.
5. Verified that the Home page loaded correctly.
6. Checked every navigation menu item.
7. Opened each subject page and confirmed that the content displayed correctly.
8. Opened every programming language page.
9. Verified all **Learn More** and **Explore** buttons.
10. Tested page layout consistency across all webpages.

11. Checked the website in **Microsoft Edge** for browser compatibility.
 12. Verified responsive behavior by resizing the browser window.
 13. Corrected any broken links and layout issues before final deployment.
-

14. 6.3 Performance Outcome

The testing results indicate that **CS Nexus** successfully meets the intended functional requirements.

The website demonstrated the following outcomes:

- All webpages loaded successfully without errors.
- Navigation links functioned correctly across the entire website.
- Subject pages and programming language pages opened successfully.
- The user interface remained consistent on every webpage.
- The website loaded quickly because it uses only HTML and CSS.
- No broken hyperlinks were found after final testing.
- The website displayed correctly in modern web browsers.
- The responsive layout provided a satisfactory viewing experience on different screen sizes.

Overall, the performance testing confirmed that the website is stable, user-friendly, lightweight, and suitable for educational purposes. The project successfully achieved its objective of providing an organized platform for learning Computer Science subjects and programming languages.

6 My learnings

The Industrial Internship provided me with valuable practical experience and significantly enhanced my technical and professional skills. Throughout the six-week internship, I worked on designing and developing **CS Nexus – A Computer Science Educational Website Using HTML & CSS**, which allowed me to apply theoretical knowledge to a real-world project.

During the development of this project, I gained a deeper understanding of **HTML5** and **CSS3** by creating a complete multi-page website with a consistent layout, responsive design, and user-friendly navigation. I learned how to organize web pages, structure content effectively, use CSS for styling, and improve the overall appearance of a website.

I also became familiar with the complete software development process, including project planning, requirement analysis, webpage design, implementation, testing, debugging, documentation, and project submission through GitHub. These activities improved my logical thinking, analytical ability, problem-solving skills, and attention to detail.

Another important learning outcome was understanding the significance of project documentation. Preparing this internship report helped me learn how to present technical work in a professional manner, explain project objectives clearly, and document every stage of software development.

In addition to technical knowledge, this internship enhanced my time management, communication, and self-learning abilities. Completing the project within the given timeline taught me the importance of planning tasks effectively and meeting deadlines.

Overall, this internship has increased my confidence in developing web applications and has motivated me to continue learning advanced technologies such as JavaScript, React, backend development, Data Science, and Machine Learning. The knowledge and experience gained during this internship will be highly beneficial for my higher studies and future career in the software industry.

7 Future work scope

The current version of **CS Nexus** is a static educational website that provides structured information about Computer Science subjects and programming languages. Although the project successfully meets its objectives, several enhancements can be implemented in future versions to improve functionality, user experience, and learning effectiveness.

Some possible future improvements include:

- Implementing **JavaScript** to add interactive features such as quizzes, animations, and dynamic content.
- Developing a **user login and registration system** to provide personalized learning experiences.
- Integrating a **backend database** using technologies such as MySQL or MongoDB to store user information, course details, and progress.
- Adding a **search functionality** to help users quickly find subjects, programming languages, and learning materials.
- Including **video tutorials**, coding examples, and downloadable study materials for each topic.
- Introducing **online assessments and quizzes** with instant feedback to help learners evaluate their understanding.
- Adding a **dark mode** and customizable themes to improve accessibility and user comfort.
- Making the website fully **responsive** for mobile phones and tablets to provide a better experience on different devices.
- Hosting the website on a public web server so that it can be accessed online by students from anywhere.
- Expanding the content by including additional Computer Science subjects, Data Science topics, Machine Learning concepts, interview preparation materials, and programming challenges.

These future enhancements will transform CS Nexus from a static educational website into a more interactive and comprehensive learning platform that can benefit a larger number of students and aspiring software developers.

8 Conclusion

The **CS Nexus – A Computer Science Educational Website Using HTML & CSS** project was successfully completed during the six-week Industrial Internship conducted by **upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)**. The project fulfilled its objective of creating an organized, user-friendly, and informative platform for learning fundamental Computer Science subjects and programming languages.

Through this internship, I gained practical knowledge in web development, project planning, testing, documentation, and GitHub version control. I also improved my problem-solving, analytical thinking, and project management skills. This experience has strengthened my technical foundation and increased my confidence in developing real-world software applications.

I believe the knowledge and skills gained during this internship will serve as a strong foundation for my future studies and career in the field of **Data Science, Machine Learning, and Software Development**.